

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
Department of Mechanical Engineering

LAB MANUAL

OPTIMIZATION TECHNIQUES

(ME 420)

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ME – 420 OPTIMIZATION TECHNIQUES	
CO1	Understand basics and engineering applications of optimization
CO2	Get familiar with mathematical software to solve optimization problems in MATLAB
CO3	Understand basics of Classical Optimization Techniques
CO4	Understand Elimination Methods of Unconstrained Optimization
CO5	Understand Interpolation Methods of Unconstrained Optimization
CO6	Understand Direct Root Methods of Unconstrained Optimization
CO7	Understand and solve Equality Constraints problems of Constrained Optimization
CO8	Understand and solve Inequality Constraints problems of Constrained Optimization

List of Experiment (ME420-OT)		
Sr. No.	Title	Course Outcomes
1	Introduction to Optimization	CO1
2	Introduction to MATLAB	CO2
3	Classical Optimization Techniques	CO3
4	Unconstrained Optimization: Elimination Methods	CO4
5	Unconstrained Optimization: Interpolation Method	CO5
6	Unconstrained Optimization: Direct Root Methods	CO6
7	Constrained Optimization: Equality Constraints	CO7
8	Constrained Optimization: Inequality Constraints	CO8

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List of Assignment (ME420-OT)		
Sr. No.	Title	Course Outcomes
1	Assignment 1- Questions on basics of Optimization	CO1
2	Assignment 2- Problems on Elimination Methods of Unconstrained Optimization	CO4
3	Assignment 3- Problems on Direct Root Methods of Unconstrained Optimization	CO6
4	Assignment 4- Problems on Constrained Optimization: Equality & Inequality Constraints	CO7, CO8

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Optimization Techniques (ME420)

Date:

Experiment No 1 **Introduction to Optimization Techniques**

AIM

Introduction to optimization techniques.

INTRODUCTION

Optimization is the act of obtaining the best result under given circumstances. In design, construction, and maintenance of any engineering system, engineers have to make technological and managerial decisions at several stages. The ultimate goal of all such decisions is either to minimize the effort required or to maximize the desired benefit. Since the effort required or the benefit desired in any practical situation can be expressed as a function of certain decision variables, optimization can be defined as the process of finding the conditions that give the maximum or minimum value of a function.

ENGINEERING APPLICATIONS OF OPTIMIZATION

Optimization, in its broadest sense, can be applied to solve any engineering problem. Some typical applications from different engineering disciplines indicate the wide scope of the subject:

1. Design of aircraft and aerospace structures for minimum weight
2. Finding the optimal trajectories of space vehicles
3. Design of civil engineering structures such as frames, foundations, bridges, towers, chimneys, and dams for minimum cost
4. Minimum-weight design of structures for earthquake, wind, and other types of random loading
5. Design of water resources systems for maximum benefit
6. Optimal plastic design of structures
7. Optimum design of linkages, cams, gears, machine tools, and other mechanical components
8. Selection of machining conditions in metal-cutting processes for minimum production cost
9. Design of material handling equipment, such as conveyors, trucks, and cranes, for minimum cost
10. Design of pumps, turbines, and heat transfer equipment for maximum efficiency
11. Optimum design of electrical machinery such as motors, generators, and transformers
12. Optimum design of electrical networks
13. Shortest route taken by a salesperson visiting various cities during one tour
14. Optimal production planning, controlling, and scheduling
15. Analysis of statistical data and building empirical models from experimental results to obtain the most accurate representation of the physical phenomenon
16. Optimum design of chemical processing equipment and plants

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17. Design of optimum pipeline networks for process industries
18. Selection of a site for an industry
19. Planning of maintenance and replacement of equipment to reduce operating costs
20. Inventory control
21. Allocation of resources or services among several activities to maximize the benefit
22. Controlling the waiting and idle times and queuing in production lines to reduce the costs
23. Planning the best strategy to obtain maximum profit in the presence of a competitor
24. Optimum design of control systems

ELEMENTS OF PROBLEM FORMULATION

(i) *Design Vector*

Any engineering system or component is defined by a set of quantities some of which are viewed as variables during the design process. In general, certain quantities are usually fixed at the outset and these are called *pre-assigned parameters*. All the other quantities are treated as variables in the design process and are called *design* or *decision variables* x_i , where $i = 1, 2, \dots, n$. The design variables are collectively represented as a design vector $\mathbf{X} = \{x_1, x_2, \dots, x_n\}^T$.

(ii) *Design Constraints*

In many practical problems, the design variables cannot be chosen arbitrarily; rather, they have to satisfy certain specified functional and other requirements. The restrictions that must be satisfied to produce an acceptable design are collectively called *design constraints*. Constraints that represent limitations on the behavior or performance of the system are termed *behavior* or *functional constraints*. Constraints that represent physical limitations on design variables, such as availability, fabricability, and transportability, are known as *geometric* or *side constraints*.

(iii) *Constraint Surface*

To visualize a constrained surface, consider an optimization problem with only inequality constraints $g_j(\mathbf{X}) \leq 0$. The set of values of $\mathbf{X}$ that satisfy the equation $g_j(\mathbf{X}) = 0$ forms a hypersurface in the design space and is called a *constraint surface*. Note that this is an $(n-1)$ dimensional subspace, where n is the number of design variables. The constraint surface divides the design space into two regions: one in which $g_j(\mathbf{X}) < 0$ and the other in which $g_j(\mathbf{X}) > 0$. Thus the points lying on the hypersurface will satisfy the constraint $g_j(\mathbf{X})$ critically, whereas the points lying in the region where $g_j(\mathbf{X}) > 0$ are infeasible or unacceptable, and the points lying in the region where $g_j(\mathbf{X}) < 0$ are feasible or acceptable. The collection of all the constraint surfaces $g_j(\mathbf{X}) = 0, j = 1, 2, \dots, m$, which separates the acceptable region is called the *composite constraint surface*.

Fig 1.1 shows a hypothetical two-dimensional design space where the infeasible region is indicated by hatched lines. A design point that lies on one or more than one constraint surface is called a *bound point*, and the associated constraint is called an *active constraint*. Design points that do not lie on any constraint surface are known as *free points*. Depending on whether a

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particular design point belongs to the acceptable or unacceptable region, it can be identified as one of the following four types:

1. Free and acceptable point
2. Free and unacceptable point
3. Bound and acceptable point
4. Bound and unacceptable point

All four types of points are shown in *Fig. 1.1*.

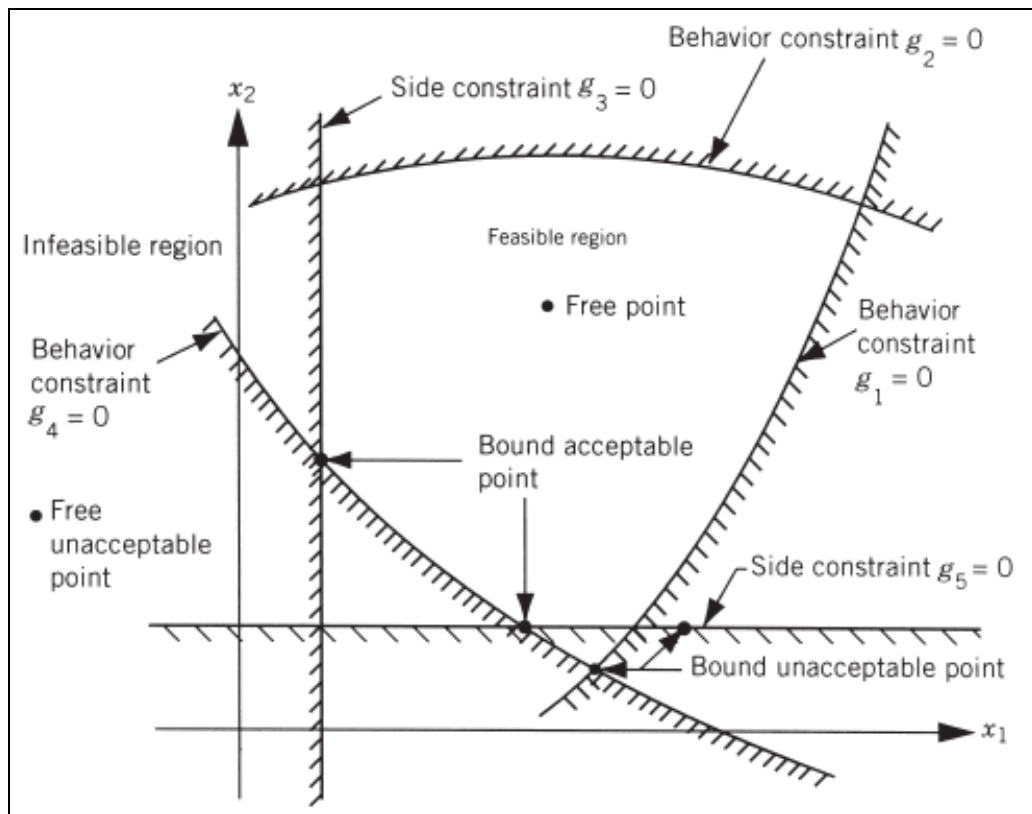


Fig 1.1: Constraint surface in a hypothetical 2-dimensional space

(iv) Objective Function

The conventional design procedures aim at finding an acceptable or adequate design that merely satisfies the functional and other requirements of the problem. In general, there will be more than one acceptable design, and the purpose of optimization is to choose the best one of the many acceptable designs available. Thus a criterion has to be chosen for comparing the different alternative acceptable designs and for selecting the best one. The criterion, with respect to which the design is optimized, when expressed as a function of the design variables, is known as the *criterion or merit or objective function*.

(v) Objective Function Surfaces

The locus of all points satisfying $f(\mathbf{X}) = C = \text{constant}$, forms a hypersurface in the design space, and each value of C corresponds to a different member of a family of surfaces. These surfaces, called *objective function surfaces*, are shown in a hypothetical two-dimensional design in Fig. 1.2.

space in Fig. 1.5. Once the objective function surfaces are drawn along with the constraint surfaces, the optimum point can be determined without much difficulty. But the main problem is that as the number of design variables exceeds two or three, the constraint and objective function surfaces become complex even for visualization and the problem has to be solved purely as a mathematical problem. The following example illustrates the graphical optimization procedure.

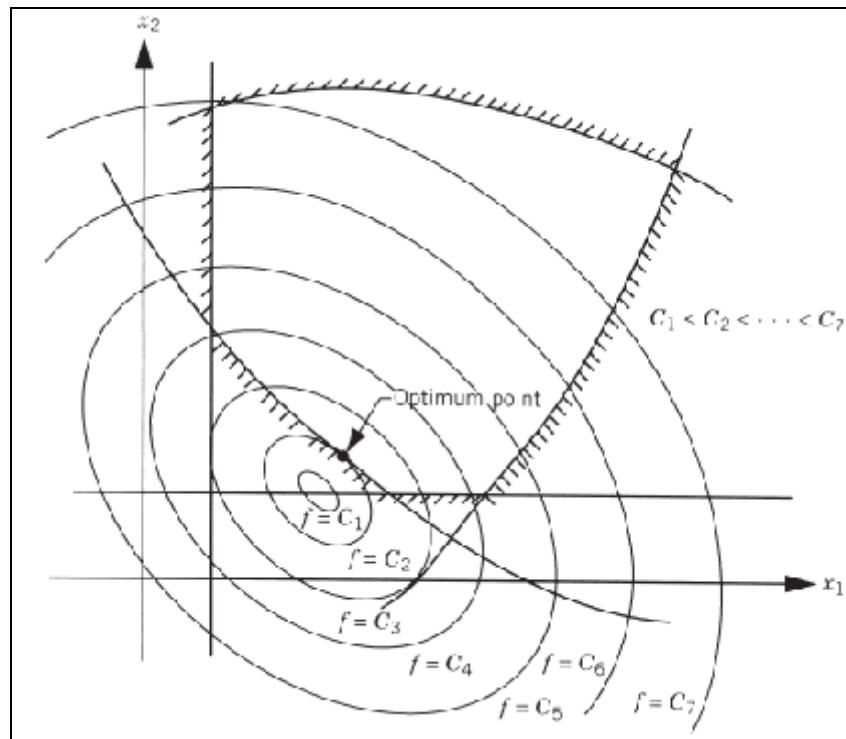


Fig 1.2: Contours of the objective function

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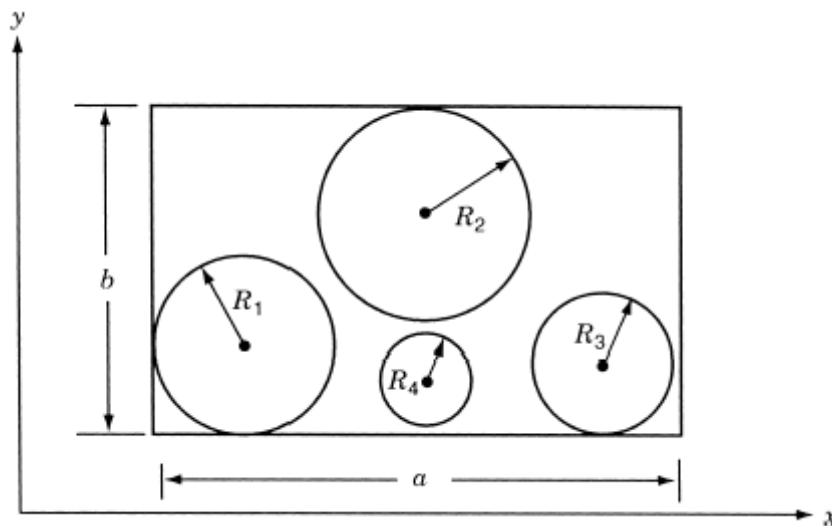
CLASSIFICATION OF OPTIMIZATION PROBLEMS

Optimization problems can be classified in several ways, as described below.

1. Classification Based on the Existence of Constraints
2. Classification Based on the Nature of the Design Variables
3. Classification Based on the Physical Structure of the Problem
4. Classification Based on the Nature of the Equations Involved
5. Classification Based on the Permissible Values of the Design Variables
6. Classification Based on the Deterministic Nature of the Variables
7. Classification Based on the Separability of the Functions
8. Classification Based on the Number of Objective Functions

EXERCISES

1. What is the difference between a constraint surface and a composite constraint surface?
2. What is the difference between a bound point and a free point in the design space?
3. It is required to stamp four circular disks of radii R_1 , R_2 , R_3 , and R_4 from a rectangular plate in a fabrication shop as shown figure bellow. Formulate the problem as an optimization problem to minimize the scrap. Identify the design variables, objective function, and the constraints.



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Optimization Techniques (ME420)

Date:

Experiment No 2 **Introduction to MATLAB**

AIM

Introduction to MATLAB.

INTRODUCTION

MATLAB stand for MATrix LABoratory is a high performance language for technical computing. It integrates computation, visualization and programming in easy to use environment. It includes:

- Modeling
- Simulation
- Data analysis
- Scientific and Engineering Graphics

EXERCISES

1. Explain the following functions

- 1) linspace
- 2) fminbnd
- 3) fminunc
- 4) linprog
- 5) quadprog
- 6) fmincon
- 7) fminimax

2. A Company produced two types of toys: P and Q. Production time of Q is twice that of P and company has maximum of 2000 time units per day. The supply of raw material is just sufficient to produce 1500 toys (of any two) per day. Toy type Q requires an electric switch which available at 600 pieces per day only. The company makes a profit of Rs.3 and Rs.5 on Type P and Q respectively. For maximization of profits, find the daily production quantities of P and Q toys. (USE MATLAB)

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Optimization Techniques (ME420)

Date:

Experiment No 3 Classical Optimization Techniques

AIM

Study of classical optimization techniques.

INTRODUCTION

The classical methods of optimization are useful in finding the optimum solution of continuous and differentiable functions. These methods are analytical and make use of the techniques of differential calculus in locating the optimum points. A study of the calculus methods of optimization forms a basis for developing most of the numerical techniques for optimization.

EXERCISES

1. State the necessary and sufficient conditions for the minimum of a function of single variable and multi-variable.
2. Define a saddle point and indicate its significance.
3. A rectangular sheet of metal with sides a and b has four equal square portion (of side d) removed at the corners, and the sides are then turned up so as to form an open box. Find the expression for depth of the box that maximizes the volume.
4. Determine whether each of the matrices given below is positive definite, negative definite, or indefinite.

$$[A] = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 4 & 2 & -4 \\ 2 & 4 & -2 \\ -4 & -2 & 4 \end{bmatrix}$$

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Optimization Techniques (ME420)

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Experiment No 4
Unconstrained Optimization: Elimination Methods

AIM

Study and computer implementation of one-dimensional elimination methods to compute optimal solution.

INTRODUCTION

If the expressions for the objective function and the constraints are fairly simple in terms of the design variables, the classical methods of optimization can be used to solve the problem. On the other hand, if the optimization problem involves the objective function and/or constraints that are not stated as explicit functions of the design variables or which are too complicated to manipulate, we cannot solve it by using the classical analytical methods.

In this situation, one-dimensional minimization methods can be applied as shown in Fig. 4.1. The minimum of a function can be evaluated in two phases. First, a crude technique is used to find a lower and upper bound of the minimum (bracketing methods). Thereafter more sophisticated method is used to search within these limits and find the optimum by successively eliminating subintervals so as to reduce the interval of successive search. These methods are called region elimination methods.

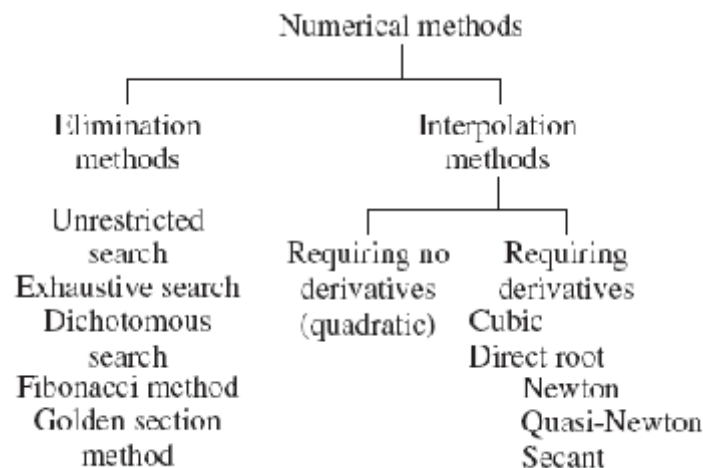


Fig 4.1: One-dimensional minimization methods

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EXERCISES

1. Find the minimum of the function

$$f = \lambda^5 - 5\lambda^3 - 20\lambda + 5$$

Using the following methods:

- Unrestricted search with a fixed step size of 0.1 from the starting point 0.0
- Unrestricted search with an accelerated step size using an initial step size of 0.1 and starting point of 0.0
- Exhaustive search method in the interval (0, 3) to achieve an accuracy of within 5% of the exact value
- Dichotomous search method in the interval (0, 3) to achieve an accuracy of within 5% of the exact value using a value of $\delta = 0.0001$
- Fibonacci search in the interval (0, 3)

2. Find the minimum of the function by the following methods:

$$F(x) = 0.65 - \frac{0.75}{1+x^2} - 0.65 \tan^{-1} \frac{1}{x}$$

- Interval halving method in the interval (0, 5) to achieve an accuracy of within 5% of the exact value
- Golden section method in the interval (0, 5)
- Enumerate solution algorithm and draw flowchart of above two methods
- Develop computer program for these methods and compare computation time (number of iterations, absolute time) for above methods

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Optimization Techniques (ME420)

Date:

Experiment No 5

Unconstrained Optimization: Interpolation Methods

AIM

Study and computer implementation of one-dimensional interpolation methods to compute optimal solution.

INTRODUCTION

The region elimination methods require only that the function be unimodal. Hence, they are applicable to both continuous and discontinuous functions as well as to discrete variables. The entire search logic in these methods is based upon a simple comparison of function values at two trial points. Moreover, this comparison only takes account of the ordering of the function values and in no way considers the difference in the magnitude of the function. The interpolation methods do take into account the relative magnitudes of the function values and as a consequence often perform better than the region elimination methods. However, interpolation methods are applicable to only smooth functions.

The simplest polynomial interpolation is a quadratic approximation. It is based upon the observation that if a function takes on its minimum in the interior of an interval, then it must be at least quadratic. If it is linear, it will assume its optimal value at one or the other of the endpoints of the interval. Thus, a quadratic estimation scheme assumes that within the bounded interval the function can be approximated by a quadratic and this approximation will improve as the points used to construct the approximation approach the actual minimum.

Cubic interpolation method is a polynomial approximation method in which a function to be minimized is approximated by a third-order polynomial. The basic logic is similar to the quadratic approximation scheme. However, in this method, because both the function value and the derivative value are derived at each point, the approximating polynomial can be constructed using fewer points.

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EXERCISES

1. Find the minimum of $f(\lambda) = \lambda^5 - 5\lambda^3 - 20\lambda + 5$ with quadratic interpolation method
2. Find the minimum of the function $f(\lambda) = \lambda / \log \lambda$ by the following methods (take the initial trial step length as 0.1)

- a) Quadratic interpolation method
- b) Powell's method
- c) Cubic interpolation method
- d) Write flowchart for cubic interpolation method

3. Minimize the function $f(x) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ starting from the point $X_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$ along the direction $S = \begin{Bmatrix} -1 \\ 0 \end{Bmatrix}$ using the quadratic interpolation method with an initial step length of 0.1.

4. Find the minimum of the following function:

$$f(\lambda) = 0.65 - \frac{0.75}{1 + \lambda^2} - 0.65 \tan^{-1} \frac{1}{\lambda}$$

Using standard MATLAB functions.

5. Find the maximum of the function given by equation:

$$f(\lambda) = \frac{0.5}{\sqrt{1 + \lambda^2}} - \sqrt{1 + \lambda^2} \left(1 - \frac{0.5}{1 + \lambda^2} \right) + \lambda$$

Using standard MATLAB function. Assume the bounds on λ as 0 and 3.

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Optimization Techniques (ME420)

Date:

Experiment No 6 Unconstrained Optimization: Direct Root Methods

AIM

Study of one-dimensional interpolation methods to compute optimal solution.

INTRODUCTION

The unconstrained optimization based on region elimination and interpolation principle assumes unimodality of the function and, in some cases, continuity of the function to be optimized. In addition to continuity, if it is assumed that the function is differentiable, then further efficiency of the optimization methods can be enhanced. When function have third-degree or higher terms involving independent variable λ , direct analytic solution would be difficult. Hence, a search method that successively approximates the stationary point of function is required.

The necessary condition for a function $f(\lambda)$ to have a minimum or maximum of λ is that $f'(\lambda) = 0$. The root (or solution) of the equation, $f'(\lambda) = 0$ can be calculated by employing iterative methods like Newton, the quasi-Newton, and the secant.

EXERCISES

1. Explain working principle of Newton and secant method with neat sketch.
2. Derive the expression for quasi-Newton formula using central difference and two point forward difference schemes.
3. Find the maximum of the following function using:
 - a. Newton method with the initial solution point 0.6
 - b. Quasi-Newton method with the initial solution point 0.6 and a finite difference step size of 0.001 (use central difference scheme).
 - c. Secant method with bounds on λ as 0 and 3

$$f(\lambda) = \frac{0.5}{\sqrt{1+\lambda^2}} - \sqrt{1+\lambda^2} \left(1 - \frac{0.5}{1+\lambda^2} \right) + \lambda$$

4. Enumerate solution algorithm and draw flowchart of Newton method

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Optimization Techniques (ME420)

Date:

Experiment No 7 Constrained Optimization: Equality Constraints

AIM

Solution approaches for the optimization problems having equality constraints.

INTRODUCTION

Most engineering problems involve optimization subject to several constraints on the design variables. The presence of constraints essentially reduces the region in which we search for the optimum. At the outset, it may appear that the reduction in the size of the feasible region should simplify the search for the optimum. On the contrary, the optimization process becomes more complicated, since optimization methods developed for unconstrained problems do not hold in the presence of constraints. In principle, this problem can be solved as an unconstrained optimization problem by explicitly satisfying the equality constraints. Method of direct substitution and Lagrange multiplier can be employed to convert constraint problem into unconstrained problem.

The method of Lagrange multipliers essentially gives a set of necessary conditions to identify candidate optimal points of equality-constrained optimization problems. This is done by converting the constrained problem to an equivalent unconstrained problem with the help of certain unspecified parameters known as Lagrange multiplier.

EXERCISES

1. Find the dimensions of a straight beam of circular cross-section that can be cut from a conical log of height h and base radius r to maximize the volume of the beam.
2. Find a , b , and c that maximizes function, $f = \frac{6abc}{a+2b+2c}$ using Lagrange multiplier method

Where a , b , and c are restricted by the relation $abc = 16$

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Optimization Techniques (ME420)

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Experiment No 8 Constrained Optimization: Inequality Constraints

AIM

Solution approaches for the optimization problems having inequality constraints.

INTRODUCTION

Lagrange multipliers can be used in developing optimality criteria for the equality constrained optimization problems. Kuhn and Tucker have extended this theory to include the general non-linear programming problem with both equality and inequality constraints.

Kuhn and Tucker have developed the necessary and sufficient optimality conditions for the non-linear programming problem assuming that the objective function and constraint functions are differentiable. These optimality conditions, commonly known as the Kuhn-Tucker conditions (KTCs) may be stated in the form of finding a solution to a system of nonlinear equations. They are also referred to as the Kuhn-Tucker problem (KTP).

EXERCISES

1. State the Kuhn-Tucker conditions and explain active and inactive constraints.

2. Consider the problem: Minimize $f = x_1^2 + x_2^2 + x_3^2$

Subjected to: $x_1 + x_2 + x_3 \geq 5$, $2 - x_2x_3 \leq 0$

$x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 2$

Determine whether the Kuhn-Tucker conditions are satisfied at the following points:

$$X_1 = \begin{Bmatrix} 3/2 \\ 3/2 \\ 2 \end{Bmatrix}, X_2 = \begin{Bmatrix} 4/3 \\ 2/3 \\ 3 \end{Bmatrix}, X_3 = \begin{Bmatrix} 2 \\ 1 \\ 2 \end{Bmatrix}$$

3. Consider the optimization problem: Maximize $f = -x_1 - x_2$

Subjected to: $x_1^2 + x_2 \geq 2$, $4 \leq x_1 + 3x_2$, $x_1 + x_2^4 \leq 30$

- a. Find whether the design vector $X = \begin{Bmatrix} 1 \\ 1 \end{Bmatrix}$ satisfy the Kuhn-Tucker conditions for a constrained problem
- b. What are the values of the Lagrange multiplier at the given design vector?

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